

- (d) The circular motion of the ball in Fig. 1.1 has a diameter of 0.46 m and an angular speed of 1.9 rad s^{-1} .

For the simple harmonic motion of the shadow of the ball in Fig. 1.1, calculate:

- (i) the amplitude

amplitude = m [1]

- (ii) the period

period = s [2]

- (iii) the maximum acceleration.

maximum acceleration = m s^{-2} [2]

- (e) On Fig. 1.1, draw, and label with the letter A, the position of the shadow on the screen when the shadow has its maximum positive acceleration. [1]

[Total: 15]

